Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Solving by Factoring and Other Methods **Date:** ___________

Foundation (Jalapeno)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring.
(A) (x + 2)(x + 3)
(B) (x - 2)(x - 3)
(C) (x + 2)(x - 3)
(D) (x - 2)(x + 3)
(E) (x + 1)(x + 4)

Q2. A spaceship is traveling at x km/h. If the ship's velocity is given by $x^2 - 7x - 18 = 0$, what are the possible speeds?
(A) 9 km/h, -2 km/h
(B) -9 km/h, 2 km/h
(C) 9 km/h, 2 km/h
(D) -9 km/h, -2 km/h
(E) 18 km/h, -1 km/h

Q3. Verify the solutions to $x^2 + 2x - 6 = 0$.
(A) x = -1 pm sqrt7
(B) x = 1 pm sqrt7
(C) x = -3, x = 2
(D) x = -2, x = 3
(E) x = 3, x = -2

Q4. A lab experiment involves a quadratic equation $x^2 - 4x - 3 = 0$. What are the values of x?
(A) x = -1, x = 3
(B) x = 1, x = -3
(C) x = -3, x = 1
(D) x = 3, x = -1
(E) x = 2, x = -2

Q5. Solve $x^2 - 9x + 20 = 0$ using factoring.
(A) (x - 4)(x - 5)
(B) (x - 5)(x - 4)
(C) (x + 4)(x + 5)
(D) (x + 5)(x + 4)
(E) (x - 2)(x - 10)

Q6. A planet's orbit can be modeled by $x^2 + 2x - 15 = 0$. What are the solutions?
(A) x = -5, x = 3
(B) x = -3, x = 5
(C) x = 5, x = -3
(D) x = 3, x = -5
(E) x = -2, x = 7

Q7. Solve $x^2 + x - 12 = 0$.
(A) (x + 4)(x - 3)
(B) (x + 3)(x - 4)
(C) (x + 1)(x - 12)
(D) (x - 1)(x + 12)
(E) (x + 2)(x - 6)

Q8. A physics experiment involves $x^2 - 2x - 15 = 0$. What are the values of x?
(A) x = -3, x = 5
(B) x = 3, x = -5
(C) x = 5, x = -3
(D) x = -5, x = 3
(E) x = 1, x = -15

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 4x + 4 = 0$ and verify the solutions. Show all steps.

Q10. A spaceship is traveling through space with a velocity given by $x^2 - 5x - 6 = 0$. Solve for x and explain the physical meaning of the solutions.

Chef's Tips

- Remind students to factor the quadratic equations carefully.
- Encourage students to verify their solutions.
- Allow students to use calculators to check their answers.